

Enderton's Elements of Set Theory

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Contents

1	Axioms and Operations	2
1.1	Axioms	2

Chapter 1

Axioms and Operations

1.1 Axioms

Extensionality axiom

If two sets have the exact same members, then they are equal:

$$\forall A \forall B [\forall x (x \in A \leftrightarrow x \in B) \rightarrow A = B]$$

(The nested implication is not an equivalence because the other direction is already true. Declaring it would be redundant.)

Empty set axiom

There is a set having no members:

$$\exists B \forall x (x \notin B)$$

Pairing axiom

For any sets u and v , there is a set having as members just u and v :

$$\forall u \forall v \exists B \forall x (x \in B \leftrightarrow x = u \text{ or } x = v)$$